higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

QUANTITY SURVEYING N5

11 APRIL 2019

This marking guideline consists of 11 pages.

Copyright reserved

SECTION A

QUESTION 1

1.1

2 x 3 000 = 6,000
3 x 220 = 0,660
6,660 √

4√ × 6,660√ = 26,640√ m

(4)

1.2

26,640√
-4/0,330 = 1,320√
25,320 m√

(3)

1.3

6,660√ – 2/0,330
= 6,000√ × 2√ =12,000√
- 0,330√
11,670 m√

(5)

(1)

1.4

Height
0,600√
+0,200√
0,800 m√

(2)

1.5

0,825 m√

(1)

[15]

(2)

QUESTION 2

2.1

26,64		Excav. for fdn. trenches
0,70	√	< 2 m from the g.l.
0,80		
11,67		
0,70	√	
0,80		

(3)

2.2

Ddt

6	√	
0,70	√	Excav√. for s.t. a.b.
0,19	√	
0,80	√	

0,700 – 0,330
= 0,370 ÷ 2 = 0,185√ m

(4)

2.3

2		R.o.c. to√ sides of fdn.
26,64		trenches < 1,5 m from.g.l.
2 0,80	√	
11,67		
0,80	√	

(3)

(3)

2.4

6	√	ddt
0,70		R.o.c. to√ sides of s.t. a.b.
6 √ 0,80	√	
2		
0,19		
0,80	√	

(4)

(4)

[14]

QUESTION 3

3.1

26,64		330 mm thick fdn. wl.
0,83	√	
11,67		
0,83		

(3)

3.2

26,64		Mass conc. in fdn ftg.
0,70	√	
0,20		
11,67		
0,70	√	
0,20		

(3)

3.3

Ddt.

6		Mass conc. √ in fdn ftg.
0,70		a.b.
0,19	√	
0,20		

(3)

3.4

2		B/fill to √ sides of fdn
26,64		wls
0,19		
2		
0,80	√	
11,67		
0,19		
0,80		

(3)

3.5

Ddt.

6	√	B/fill to √ sides of fdn
0,70		a.b.
0,19	√	
0,80		

0,330 + 2/0,185
= 0,700 m√

(4)

[16]

(5) (6)

QUESTION 4.1

1.2

2√ x 3 000√ = 6,000√
3 x 220 = 0,660√
6,660√

4√ x 6,660√ = 26,640√ m

1.2

26,640√
-4√/0,330√ = 1,320√
25,320 m√

1.3

6,660√ – 2√/0,330√
= 6,000√ x 2 =12,000√
- 0,330 √
11,670 m√

(1)

1.4

Height
0,600√
+ 0,200√
0,800 m√

1.5

0,825 m√

Marks
- Check waste calculations 1
- Check side casts 0,5
- Check dimensioning 0,5
- Square dimensions 0,5
- Check squaring 0,5
- Transfer items 0,5
- Check transfers 0,5
(4)

(2)

QUESTION 4.1

2.1

Excav. for fdn. trenches < 2 m from the g.l.

26,64√ × 0,70√ × 0,80 = 14,92 √
11,67 × 0,70 × 0,80 = 6,54
21,46

2.2

Ddt

Excav. for s.t. a.b.

6√ × 0,70√ × 0,19√ × 0,80√ = 0,64 √

0,700√ – 0,330 √
= 0,370√ ÷ 2√ = 0,185 m

2.3

R.o.c. to sides of fdn. trenches < 1,5 m from. g.l.

2√ × 26,64 √ × 0,80 = 42,62
11,67 √ × 0,80 √ = 18,67 √
61,29 √

(3)

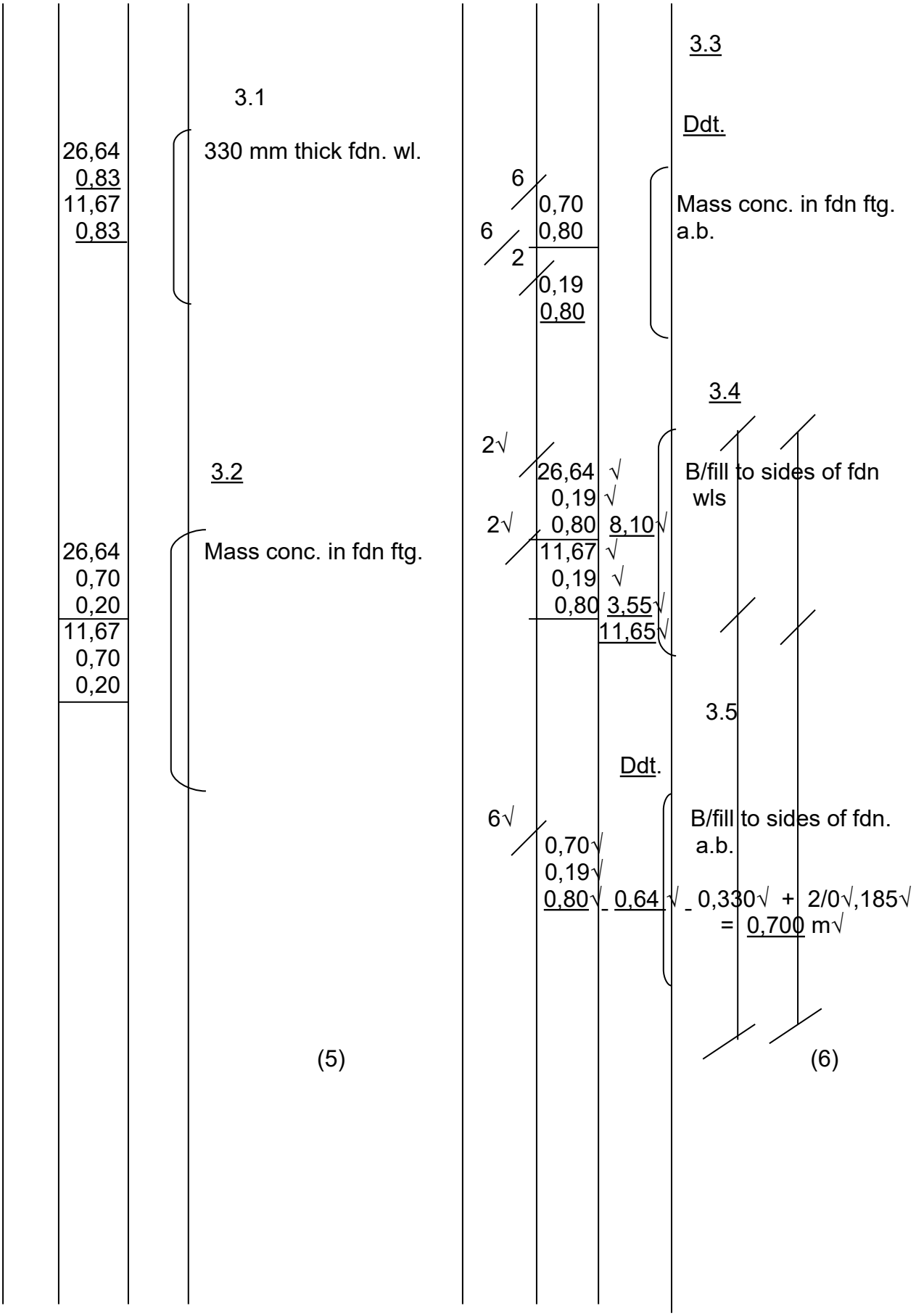
2.4

R.o.c. to sides of s.t. a.b.

6√ × 0,70 √ × 0,80 = 3,36√
2√ × 0,19√ × 0,80√ = 1,82√
5,18√

(4)

QUESTION 4.1


3.1
26,64
0,83
11,67
0,83
330 mm thick fdn. wl.
3.2
26,64
0,70
0,20
11,67
0,70
0,20
Mass conc. in fdn ftg.
(5)
3.3
Ddt.
6
0,70
6 0,80
2
0,19
0,80
Mass conc. in fdn ftg.
a.b.
3.4
2√
26,64 √
0,19 √
2√ 0,80 8,10√
11,67 √
0,19 √
0,80 3,55√
11,65√
B/fill to sides of fdn
wls
3.5
Ddt.
6√
0,70√
0,19√
0,80√ - 0,64 √ - 0,330√ + 2/0√,185√
= 0,700 m√
B/fill to sides of fdn.
a.b.
(6)

QUESTION 4.2

FOUNDATION OF MOUNTAIN HUT

EARTHWORKS

EXCAVATIONS

M³

Excav. for s.t. < 2 m dp. from g.l.

21,46	3	0,64	3
- 0,64	T		
21,00			

21 M³

M²

R.o.c. to sides of s.t. < 1,5 m from g.l.

61,29	3	5,18	4
5,18	T		
56,11			

56 M²

M³

Backfilling to sides of 330 mm fdn. wls.

11,65	6	0,64	6
- 0,64	T		
11,01			

11M³

Marks	
- Section	1
- Trade	1
- Unit	1
- Page ref.	0,5
- Reducing	0,5
- Transfer items	0,5
- Check transfers	0,5
	(5)

Copyright reserved

QUESTION 4.3

	FOUNDATION OF MOUNTAIN HUT			
	Bill 1			
	EARTHWORKS			
	EXCAVATIONS			
1	Excavate in earth for foundation trenches not exceeding 2 metres deep from the ground level.	M^3	21	
2	Risk of collapse to sides of surface trenches not exceeding 1,5 m from the ground level	M^2	56	
3	Backfilling to sides of foundation walls	M^3	11	
	Carried forward to summary			R

Marks	
Section	1
Bill no.	1
Trade	1
Item no.	0,5
Descriptions	0,5
Unit	0,5
Quantities	0,5
Summary	1
	(6)

[15]

TOTAL SECTION A: 60

Copyright reserved

SECTION B

QUESTION 5

Preliminaries	4/9 × R125 300,00	R55 690,00✓
Earthworks	100%	R99 510,00✓
Conc., Frmwrk. and Reinf.	100%	R457 680,00✓
Masonry	85% × R220 300,00	R187 260,00✓
Waterproofing	10% × R17 453,00	R1 750,00✓
Carpentry and joinery	15% × R60 443,00	R10 113,00✓
Builder's work complete		R812 003,00
Materials on site		R6 500,00✓
		R818 503,00✓
Retention	5% × R1 492 375,00	R74 620,00✓
		R743 880,00✓
Less previous payments		R682 350,00✓
Due to the contractor		R61 530,00✓

[10]

QUESTION 6

6.1 6.1.1
- Measured in m²
- Average depth required
- Place to be deposited (3)

6.1.2
- Measured in m³
- No categories
- No water
- No risk of collapse (4)

6.2 6.2.1
- They are appointed✓ by the architect✓
- to supply✓ only specialist material or units.✓

6.2.2
- They are specialist contractors✓ who perform specialist work.✓
- They are selected and appointed✓ by the architect.✓

(2 × 4) (8)
[15]

QUESTION 7

7.1
- Do not duplicate public roads.
- Do not damage the environment unnecessarily.
- Plan early for traffic control.

7.2
- Operators must have a clear view when operating.
- Station noisy plant far away from site offices.
- Consider the installation regulations and safety of workers.

7.3
- They should be near to stores to easily receive deliveries.
- Find the best area to be able to overlook the site.
- All offices should be near the gate to monitor visitors.

7.4
- Incorporate accommodation for the storekeeper.
- Allow for enough off-loading space for the delivery vehicles.
- Place them near the gate for ease delivery.

7.5
- Stripping of top soil
- Demolitions of structures
- Clearing away of rubbish, grass, shrubs and trees

(5 × 3) **[15]**

TOTAL SECTION B: 40
GRAND TOTAL: 100

Copyright reserved